Vasudha Devarakonda

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Research Summary

I am a second-year Ph.D. student at Texas A&M University working on memory-efficient ML systems. My research focuses on improving deep learning performance in multi-tenant cloud environments, with an emphasis on memory-efficient systems design.

Skills

Languages: Python, C/C++, Go, Bash, R
AI/ML: PyTorch, CUDA, TensorFlow
Cloud & Tools: Linux, AWS, Kubernetes, Docker, Git, GitHub, Slurm, LaTeX

Professional Experience

Ph.D. Student (Advisor: Dr. Khanh Nguyen)
COSMOS, Texas A&M University | January 2025–Present

- Research memory-efficient execution of ML training and inference workloads in multi-tenant cloud systems, with a focus on resource-aware memory management.
- **Ongoing Project**: Multi-tenant GPU cluster scheduling and memory management for DNN and LLM inference workloads.

Teaching Assistant
Distributed Systems and Cloud Computing, Texas A&M University | August 2026–Present

- Assist with lab sessions, projects, and knowledge checks.

Graduate Research Assistant (Advisor: Dr. Chia Che Tsai)
SysDev, Department of Computer Science and Engineering, Texas A&M University | November 2023–2024

- Researched serverless application deployment and workload aggregation and disaggregation.

Software Engineer Associate, Cloud Unit, Research and Development
Siemens Digital Industries Software | July 2021–July 2023

- Designed a multi-region Kubernetes platform that application teams adopted to deploy services across clusters.
- Automated cloud operations and incident-response workflows to reduce manual work and accelerate response.

Publications

- D. Nguyen, **V. Devarakonda**, A. Jiang, and K. Nguyen. “Bonsai: Efficient and Optimal Automatic Tensor Rematerialization for Memory-Constrained DNN Training.” ACM OOPSLA 2026.
- **V. Devarakonda**, D. Nguyen, and K. Nguyen. “GOA: Memory-Safe Oversubscription for Deep Learning Training Workloads.” IEEE/ACM JSSPP 2026.
- **V. Devarakonda**, A. Earnest, and C. C. Tsai. “SoK: Virtualization Challenges and Techniques in Serverless Computing.” ACM SOSP Workshop on Hot Topics in System Infrastructure, 2024.

Education

Texas A&M University | College Station, TX, USA
Ph.D. Student | Jan 2025–Present
M.S. Student, Computer Engineering | Aug 2023–Jan 2025 | Transferred

Visvesvaraya National Institute of Technology | Nagpur, India
B.Tech. in Electronics and Communication Engineering | Jul 2017–Jun 2021

Activities and Awards

- Awarded the Ms. Madhur Dahake Medal in B.Tech. ECE. (2021)
- VNIT basketball player; promoted to captain and led the inter-NIT team. (2017–2021)